HastaDAQ

Net labels connect everything: same name = same net. +3V3 / GND are plain labels, not power symbols.

Pimoroni Pico Plus 2 (+ RM2 on SP/CE): same 40-pin header as Pico 2 W. I2C0 = GP4/GP5 (header pins 6/7), to the IMU Qwiic-breakout header and to the 4-wire loom for the strain-gauge board. Crossed-out pins are intentionally unused.

J1
IMU via Qwiic breakout (The Pi Hut 106676)

GND	1	GND
+3V3	2	3V3
SDA	3	SDA
SCL	4	SCL

J2
GAUGE BOARD LOOM (J_I2C): GND, +5V, SDA, SCL

GND	1	GND
+5V	2	5V
SDA	3	SDA
SCL	4	SCL

Strain-gauge board: 4-wire I2C loom. Pin 2 is +5V, NOT 3V3 -- use a keyed plug, never a Qwiic cable. Pull-ups live on the ADS122C04 breakouts on that board.

U1
Pimoroni Pico Plus 2 (header pins)

	1	GP0_UART0-TX	VBUS	40	
SBUS_IN_GPIO	2	GP1_UART0-RX	VSYS	39	+5V_VSYS
GND	3	GND	GND	38	GND
	4	GP2	3V3_EN	37	
	5	GP3	3V3(OUT)	36	+3V3
SDA	6	GP4_I2C0-SDA	ADC_VREF	35	
SCL	7	GP5_I2C0-SCL	GP28	34	
GND	8	GND	AGND	33	GND
	9	GP6	GP27	32	
ENC3_CS	10	GP7	GP26	31	
ENC2_CS	11	GP8	RUN	30	
GND	12	GP9	GP22	29	
SPI_SCK	13	GND	GND	28	GND
SPI_MOSI	14	GP10_SPI1-SCK	GP21_UART1-RX	27	SBUS
SPI_MISO	15	GP11_SPI1-TX	GP20_UART1-TX	26	SBUS
ENC1_CS	16	GP12_SPI1-RX	GP19	25	SSI2_DATA
GND	17	GP13_SPI1-CSn	GP18	24	SSI1_DATA
TRIG_IN_GPIO	18	GND	GND	23	GND
TRIG_OUT_GPIO	19	GP14	GP17	22	SSI_CLK
	20	GP15	GP16	21	SOFTSTOP_IN

J3
ENC1 SPI port: MT6835 (pitch)

GND	1	GND
ENC1_CS	2	CS
SPI_SCK	3	SCK
SPI_MOSI	4	MOSI
SPI_MISO	5	MISO
+3V3	6	3V3
+5V	7	5V
GND	8	GND

SSI (RLS RM08 via an inline LDB02 near the Pico; this board has the 3.3 V single-ended side only).
One shared clock SSI_CLK (GP17) drives both encoders; DATA pins GP18 / GP19 are consecutive for one PIO read.

J6
SSI1: RM08 encoder (3V3, GND, CLK, DATA)

+3V3	1	3V3
GND	2	GND
SSI_CLK	3	CLK
SSI1_DATA	4	DATA

J4
ENC2 SPI port: MT6835

GND	1	GND
ENC2_CS	2	CS
SPI_SCK	3	SCK
SPI_MOSI	4	MOSI
SPI_MISO	5	MISO
+3V3	6	3V3
+5V	7	5V
GND	8	GND

J11
SSI2: RM08 encoder (3V3, GND, CLK, DATA)

+3V3	1	3V3
GND	2	GND
SSI_CLK	3	CLK
SSI2_DATA	4	DATA

Pull-ups (10k): ENC CS x3, SSI1 DATA, SSI2 DATA

J5
ENC3 SPI port: RLS Orbis (or MT6835)

GND	1	GND
ENC3_CS	2	CS
SPI_SCK	3	SCK
SPI_MOSI	4	MOSI
SPI_MISO	5	MISO
+3V3	6	3V3
+5V	7	5V
GND	8	GND

R1 10k (+3V3 → ENC1_CS)
R2 10k (+3V3 → ENC2_CS)
R3 10k (+3V3 → ENC3_CS)
R4 10k (+3V3 → SSI1_DATA)
R15 10k (+3V3 → SSI2_DATA)

U2
Pololu D42V55F8 8.4V 5A buck (servo rail)

VSERVO	1	VOUT	VOUT	2	VSERVO
GND	3	GND	GND	4	GND
GND	5	GND	GND	6	GND
+12V	7	VIN	VIN	8	+12V
	9	VRP	VRP	10	
	11	PG	EN	12	

U3
SparkFun BabyBuck 5V (AP63357) (Pico VSYS, Orbis)

+12V	3	VIN	VOUT_5V	1	+5V
GND	2	GND			
	4	EN			

D1
Schottky (e.g. 1N5822, 3A)
+5V → A, K → +5V_VSYS

C3
100uF 10V 5V bulk (RS PRO LHK 5x7)
+5V / GND

C1
100uF 25V input bulk (Panasonic ECA1EM101)
+12V / GND

C2
470uF 16V servo bulk (Panasonic EEUFR1C471)
VSERVO / GND

J7
SBUS servo header 3x6 (6 plugs: GND, V, SIG)

GND	1	S1_GND
VSERVO	2	S1_V
SBUS	3	S1_SIG
GND	4	S2_GND
VSERVO	5	S2_V
SBUS	6	S2_SIG
GND	7	S3_GND
VSERVO	8	S3_V
SBUS	9	S3_SIG
GND	10	S4_GND
VSERVO	11	S4_V
SBUS	12	S4_SIG
GND	13	S5_GND
VSERVO	14	S5_V
SBUS	15	S5_SIG
GND	16	S6_GND
VSERVO	17	S6_V
SBUS	18	S6_SIG

TEST POINTS (plain 2.0 mm pads): two GND pads for a scope clip (near the supply and near the Pico). Rails and signals are probed at component leads and header pins (C1/C2/C3, D1, Pico pins 36/39).

GND → TP1 (TP)
GND → TP2 (TP)

PWR_FLAG: marks nets supplied through the supply connector (+12V, GND) and the Schottky (+5V_VSYS)

#FLG1 PWR_FLAG → +12V
#FLG2 PWR_FLAG → GND
#FLG3 PWR_FLAG → +5V_VSYS

J10
SBUS IN (RC receiver): GND, V, SIG

GND	1	GND
+5V	2	V
SBUS_IN	3	SIG

SBUS_IN → R13 1k series → SBUS_IN_GPIO
SBUS_IN_GPIO → R14 10k pull-down (DNP at first) → GND
D4 5.1V zener (BZX55C5V1) SBUS IN clamp: GND → A, K → SBUS_IN_GPIO

SBUS IN: RC-receiver pass-through (plug order GND, V, SIG). V = +5V (NOT VSERVO; the Futaba R3008SB takes 4.8–7.4 V, so 8.4 V would overstress it; check any other receiver's range). SIG -> 1k -> GP1 (UART0 RX); 5.1 V zener clamps a receiver that outputs more than 5.1 V (about 100 pF at 100 kbaud; harmless). 10k pull-down R: leave UNFITTED on the first trial; RP2350 erratum E9 can hold a pin with a weak pull-down (>~8.2k) near 2 V, so if it is fitted and misbehaves use 4.7k.

J8
Main supply cable (2x4): VSUPPLY, VGND, TRIG, SOFTSTOP

+12V	1	VSUPPLY
GND	2	VGND
TRIG_IN	3	TRIG_IN
SOFTSTOP_CABLE	4	SOFTSTOP+
GND	5	SOFTSTOP-
TRIG_OUT	6	TRIG_OUT
GND	7	VGND
+12V	8	VSUPPLY

TRIGGER IN: any 3.3 V .. 24 V level, opto input

TRIG_IN → R7 10k pull-down → GND
TRIG_IN → R8 1k 1W → TRIG_IN_LED
U4 FOD817A opto (TRIG_IN): A ← TRIG_IN_LED, K ← GND; C → TRIG_IN_GPIO, E → GND
+3V3 → R9 10k pull-up → TRIG_IN_GPIO
D3 1N4148 (LED reverse protect): GND → A, K → TRIG_IN_LED

R5 1k (soft-stop series): SOFTSTOP_CABLE → SOFTSTOP_IN
R6 10k (soft-stop pull-up): +3V3 → SOFTSTOP_IN
D2 5.1V zener (BZX55C5V1) soft-stop clamp: GND → A, K → SOFTSTOP_IN

TRIGGER OUT: open-collector NPN, pull-up rail chosen by jumper

TRIG_OUT_GPIO → R10 2.2k (base) → TRIG_OUT_BASE
TRIG_OUT_BASE → R11 10k (base-emitter) → GND
Q1 2N3904 NPN (TO-92, pin 1 E, 2 B, 3 C): B ← TRIG_OUT_BASE; C → TRIG_OUT; E → GND
TRIG_PU → R12 1.5k pull-up → TRIG_OUT
J9 TRIG_OUT pull-up (1x5): 3V3, PU, 5V, PU, 12V

Shunt pins 1–2 (3V3), 2–3 (5V) or 4–5 (12V) to pull TRIG_OUT up to that rail; fit none if the receiver has its own pull-up.
Pins 2 and 4 are both TRIG_PU. Fit ONE shunt only (two shunts could join rails).
Pull-up current: 3.3 V ~2 mA, 5 V ~3 mA, 12 V ~7 mA (enough for a 6.5 mA opto input such as MPS4200).
24 V receivers: no jumper, add an external 24 V pull-up in the harness. 3V3 for eSync 2 high-Z ports (3.3 V max).

NOTES
1. ONE common GND (Pico, sensors, servo PSU). SBUS needs a shared ground reference.
   RM2 WiFi breakout (off-board, cabled to the Pico Plus 2 SP/CE connector): uses GP32–GP36 only, no overlap with I2C (GP4/5) or any other signal.
2. Strain gauges: SEPARATE board (kicad/gaugeboard), reached by a 4-wire I2C loom (GND, +5V, SDA, SCL; 0.1in headers, keyed plugs).
   That board makes its own clean 3.3 V from +5V and holds both ADS122C04s, the gauge headers and the filtering.
   ADS #2 needs its ADDR0 jumper bridged (expected address 0x41, unconfirmed; firmware scans at boot). I2C pull-ups come from the ADS122C04 breakouts on the gauge board and from the IMU breakout.
3. SBUS: GP20 (TX) and GP21 (RX) on one net (UART inverted in firmware). One 3x6 servo header (6 plugs of GND, V, SIG; footprint HastaDAQ:PinHeader_3x06): V = VSERVO (8.4 V), SIG = SBUS (all six in parallel), GND.
4. POWER: 12 V in, MAX 20 V (C1 is rated 25 V; silkscreen this at JB) (no fuse: the bench supply's current limit is the protection) -> +12V rail -> (a) Pololu D42V55F8 = VSERVO 8.4 V (servos), (b) SparkFun BabyBuck 5V = +5V (Orbis, gauge-board loom, Schottky -> Pico VSYS). VSYS max is 5.5 V, USB (VBUS) is left for programming; with USB only, +5V is not live.
   BabyBuck EN is left open (its own divider enables it). 5V bulk: module has 2x22uF; 100uF added near the Pico/ports.
5. Triggers: TRIG_IN accepts 3.3–24 V (opto), TRIG_OUT is open collector with a jumper-selected pull-up. Pin order: see each symbol.
   Main supply cable (2x4, pins 1–8 row-major): 1.8 VSUPPLY (12 V) | 2.7 VGND | 3 TRIG_IN | 6 TRIG_OUT | 4 SOFTSTOP+ | 5 SOFTSTOP-.
   SOFTSTOP+ -> 1k -> GP16 (10k pull-up to 3V3, 5.1 V zener to GND). SOFTSTOP- = GND: loop closed = run, broken = stop.
   The 2x4 connectors (this one and ENC1–3) are plain 0.1in headers for now; use keyed/shrouded plugs: a rotated plug would put 5V on a CS pin. Measured system current was <= 1 A at 8.4 V well inside ~3 A per 0.1in pin.
6. Encoders: 3 x 2x4 SPI headers (pins 1–8: GND, CS, SCK, MOSI, MISO, 3V3, 5V, GND) on one SPI1 bus: CS = GP13/GP9/GP8.
   Intended: ENC1 + ENC2 = MT6835, ENC3 = RLS Orbis. MT6835 uses 3V3 (3.3 V logic); Orbis uses 5V with 3.3 V logic. Check MISO tri-states when CS is high.
   The three ENC headers are a generic SPI port (SPI1: SCK GP10, MOSI GP11, MISO GP12, own CS each, 3V3 and 5V supply), so any 3.3 V SPI device can be plugged in, not just encoders. Each device needs its own CS and an MISO that tri-states when CS is high.
7. SSI encoders: two 1x4 headers SSI1 (J6) and SSI2 (J11), pins 1–4 = 3V3, GND, CLK, DATA, for RLS RM08 encoders through an inline LDB02.
   SSI_CLK GP17 (shared), SSI1_DATA GP18, SSI2_DATA GP19, each DATA with a 10k pull-up to 3V3 (unplugged = all ones = invalid frame).

HastaArduino
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